$$\pi' \longrightarrow_{\mathsf{cut}} \dfrac{\dfrac{\overset{\pi_1^L}{\vdash F\{y/x\}, \Gamma, I_1}}{\vdash \exists x F, \Gamma, I_1}\,(\exists) \qquad \vdash I_1{}^{\perp}, \Delta}{\vdash \exists x F, \Gamma, \Delta}\,(\mathsf{Cut})$$

$$\longrightarrow_{\mathsf{cut}} \dfrac{\overset{\pi_1^L}{\vdash F\{y/x\}, \Gamma, I_1} \qquad \overset{\pi_1^R}{\vdash I_1{}^{\perp}, \Delta}}{\dfrac{\vdash F\{y/x\}, \Gamma, \Delta}{\vdash \exists x F, \Gamma, \Delta}\,(\exists)}\,(\mathsf{Cut}) \quad = \pi.$$